MANAS RANJAN DAS

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SUMMARY

Electrical & Computer Engineering student with 3+ years of hands-on experience in **AI/ML, Computer Vision, NLP and Full-Stack AI** systems. Built 3+ AI/ML applications, including predictive analytics, YOLO-based vision systems and ML browser extensions. Experienced with **Python, TensorFlow, Scikit-learn, OpenCV, NLP, MongoDB and Agentic AI**, with a focus on deployable and scalable solutions.

EDUCATION

Biju Patnaik University of Technology, Rourkela **2023 — 2027**
Bachelor of Technology in Electrical & Computer Engineering *CGPA: 8.88 / 10.00*

Jatiya Kabi Bira Kishore Das Govt. HSS **2021 — 2023**
Higher Secondary Science (CHSE Odisha) *87.5%*

Saraswati Shishu Vidya Mandir, Sharada Nagar **2019 — 2021**
High School (BSE, Odisha) *86.33%*

TECHNICAL SKILLS

Programming:	Python, C, C++, JavaScript, SQL
Machine Learning:	Scikit-learn, XGBoost, Random Forest, Regression, Predictive Analytics, SHAP, RAG
Deep Learning:	TensorFlow, YOLO, Computer Vision, OpenCV, Mediapipe
AI/NLP:	NLP, LangChain, Agentic AI, OCR, AI Applications, Hugging Face, Transformer Pipelines
Data:	Pandas, NumPy, Power BI, Streamlit, Matplotlib, Seaborn
Databases:	MongoDB, Oracle DB AI, SQLite
Web/Deployment:	React, Node.js, Express.js, Flask, Docker
Tools:	Git, GitHub, Linux, Antigravity, Codex, GitHub-Copilot, Blackbox
Soft Skills:	Leadership, Communication, Team Management, Prompt Engineering

WORK EXPERIENCE

AI Engineering Intern | Odisha Computer Application Centre (OCAC) **May 2025 — July 2025**

- Developed 3+ AI/ML applications spanning computer vision, predictive analytics and browser-based ML.
- Built a YOLO-based PCB component detector and 2+ OpenCV/MediaPipe gesture applications.
- Developed a retail forecasting pipeline using 3+ ML models with an interactive Streamlit dashboard.
- Built 2 ML-powered browser extensions for email-spam filtering and text-based emotion detection.

Cybersecurity Intern | Central Tool Room & Training Centre (CTTC) **May 2026 – Jun 2026**

- Completed hands-on training across 5+ cybersecurity domains, including network security and ethical hacking.
- Performed reconnaissance, vulnerability scanning and security analysis using Kali Linux.
- Simulated 4+ network-security mechanisms, including VPN, Port Security, Firewall and IDS/IPS.

TOP PROJECTS

Hyperion IDE — Code Security Analysis & Development Environment | ONGOING **2026**

React.js, Node.js, Express.js, MongoDB, Python NLP, Microservices, Agentic AI

- Developing an AI-assisted IDE integrating 5+ technology layers for intelligent code analysis.
- Designed a microservices architecture combining MERN + Python NLP + Agentic AI.
- Developing automated source-code vulnerability detection and intelligent developer workflows.

AI Interview Assistant — AI-Powered Interview Simulation Browser Extension **2026**

JavaScript, Node.js, YOLO, Flask-Python-OpenCV, OCR, Browser Extension

- Integrated 4+ AI/CV technologies for automated interview assistance.
- Implemented resume extraction and job-description processing for personalized interview contexts.
- Built a 100-point interview scoring system with historical performance tracking and trend visualization.

M.I.R.A — Multipurpose Intelligence for Research & Automation | ONGOING **2026**

MERN, LangChain, Agentic AI, NLP

- Developing an AI research assistant integrating 4 core technology layers.
- Built task-oriented workflows for automated research and intelligent information processing.
- Designing a scalable full-stack architecture with dedicated AI/NLP components.

SAGAR2027 — International Conference Website & its Chat Assistant **2026**

React, JavaScript, RAG, NLP, JSON, Vercel

- Developed and deployed a responsive conference platform for the 1st International Conference on Soft Computing Applications in Smart Grid & Renewable Energy.
- Built a RAG-based chatbot to provide context-aware answers from 13+ structured conference data sources.
- Integrated 13+ knowledge sources, covering schedules, registration, papers, venue, speakers, awards, accommodation and conference information.
- Implemented text + voice interaction, Markdown responses, contextual links and structured UI components.
- Deployed the production website with an integrated AI assistant for automated conference information retrieval.

Retail Sales Forecasting & Analytics **2025**

Python, Scikit-learn, XGBoost, Random Forest, Pandas, Streamlit, SHAP

- Built an ML pipeline for historical sales analysis and future demand prediction.
- Compared 3+ regression models: Linear Regression, Random Forest [97% Accuracy] and XGBoost [~94% Accuracy].
- Integrated SHAP explainability and a Streamlit dashboard for interactive model analysis.

AI/Computer Vision Applications **2025**

YOLO, OpenCV, MediaPipe, Python

- Developed a YOLO-based system for PCB component detection.
- Built 2+ gesture-controlled applications, including a writing pad and on-screen keyboard.
- Applied computer vision pipelines for real-time object and gesture recognition.

CERTIFICATIONS

- **Workshop Participation Certificate** — NxtWave (2024) : AI for Students: Built My Own GenAI Model.
- **Building AI powered ChatBot without Programming** — IBM (2024) : Built a RAG based Chat Assistant Over the IBM Watson Assistant Core Engine deployed via WordPress.
- **PYTHON FOR AI, DATA SCIENCE & DEVELOPMENT** — IBM (2025) : Hands on experience with Python on NumPy, Data Handling with Pandas, Matplotlib & Seaborn using Jupyter/Google Colab Notebook, Machine Learning.
- **Power BI Micro Cousre** — SkillCourse (2026)

ACHIEVEMENTS & LEADERSHIP

- **BPUT Tech Carnival 2025:** State-Level Silver Medal — Math Olympiad; Bronze Medal — Technical Quiz.
- **NIUS 2024 Scholar:** Selected among 71 students nationwide for NIUS Physics Program, HBCSE-TIFR.
- **Research Publication:** Co-authored research on optimizing fuzzy power system stabilizers.
- **ISTE Member:** Indian Society of Technical Education, 2024–2028.

CORE AI/ML AREAS

- Python | Machine Learning | Deep Learning | Computer Vision | NLP | Predictive Analytics | TensorFlow | Scikit-learn | YOLO | OpenCV | XGBoost | SHAP | Agentic AI | LangChain | MongoDB | SQL | RAG